US Accidents EDA Portfolio

Final Report

Generated on 2026-03-20 21:57 from the cleaned parquet, engineered features, saved model metadata, and report figures already present in this repository.

Records	**7,728,394**
Window	**Jan 2016 - Mar 2023**
Target Rate	**19.46%**
Best Model	**XGBoost**

Key Findings

- Severity 2 dominates the dataset at 79.67%, while the modeling target (Severity 3-4) accounts for 19.46%.

- Accidents cluster around commute windows, with the highest hourly volumes at 7:00 and 16:00.

- Friday has the highest accident count, and 69.33% of incidents occur during daylight.

- Visibility is constant at 10.0 miles after cleaning, so it contributes little usable variation to downstream analysis.

- California, Florida, and Texas lead total volume, with XGBoost delivering the strongest held-out performance (test AUC 0.9258).

Primary Caveats

- All road-infrastructure flags were removed before feature engineering, leaving `road_feature_count` constant at 0 for every row.

- The 2023 slice only covers data through March 31, 2023, so year-over-year totals for 2023 are not directly comparable.

Dataset Summary

Severity balance, year coverage, and top geographic concentrations from the cleaned dataset.

Severity Distribution

Severity	Accidents	Share
1	67,366	0.87%
2	6,156,981	79.67%
3	1,299,337	16.81%
4	204,710	2.65%

Year Counts

Year	Accidents
2,016	410,821
2,017	718,093
2,018	893,426
2,019	954,303
2,020	1,178,913
2,021	1,563,753
2,022	1,762,452
2,023	246,633

Top States

State	Accidents
CA	1,741,433
FL	880,192
TX	582,837
SC	382,557
NY	347,960
NC	338,199
VA	303,301
PA	296,620

Model Comparison

Held-out metrics come from models/test_metrics.json, with training time and validation AUC from training_summary.json.

Held-Out Metrics

Model	Test AUC	Test AP	Test F1	Test Accuracy	Validation AUC	Train Time (s)
XGBoost	0.9258	0.7531	0.8600	0.8506	0.9256	762.01
LightGBM	0.9171	0.7263	0.8280	0.8120	0.9167	271.57
Random Fores	0.8953	0.6697	0.7880	0.7652	0.8948	4397.04
Logistic Regres	0.7752	0.3929	0.7179	0.6861	0.7732	22.88

Top Cities

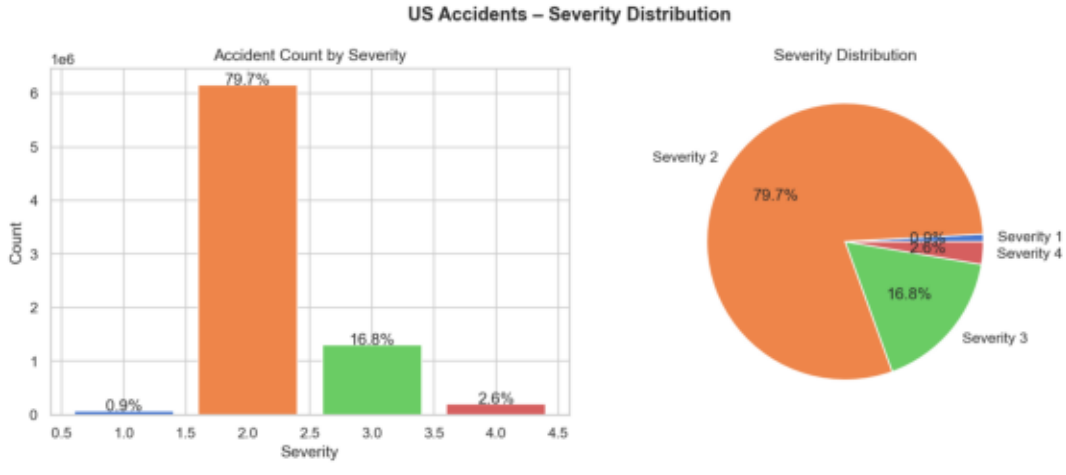
City	Accidents
Miami, FL	186,801
Houston, TX	169,428
Los Angeles, CA	156,491
Charlotte, NC	138,345
Dallas, TX	130,303
Orlando, FL	109,690
Austin, TX	96,604
Raleigh, NC	86,066

Weather Rates

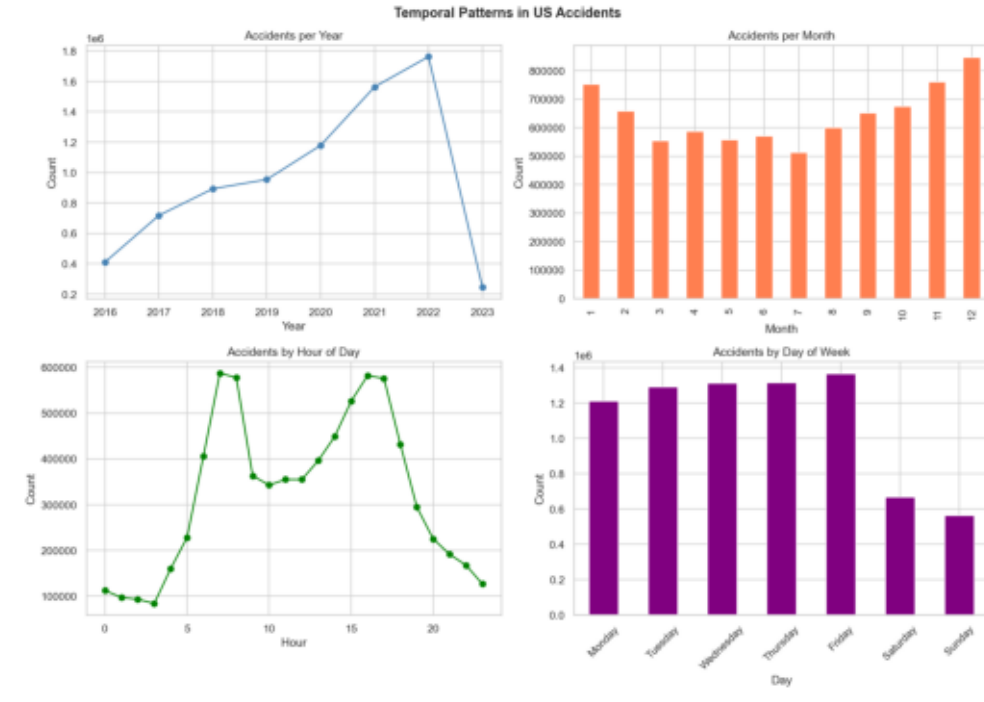
Weather Category	High-Severity Rate
Rain/Storm	22.84%
Cloudy	21.65%
Snow/Ice	19.52%
Clear	17.14%
Fog/Haze	16.19%
Other	10.61%

EDA Figures

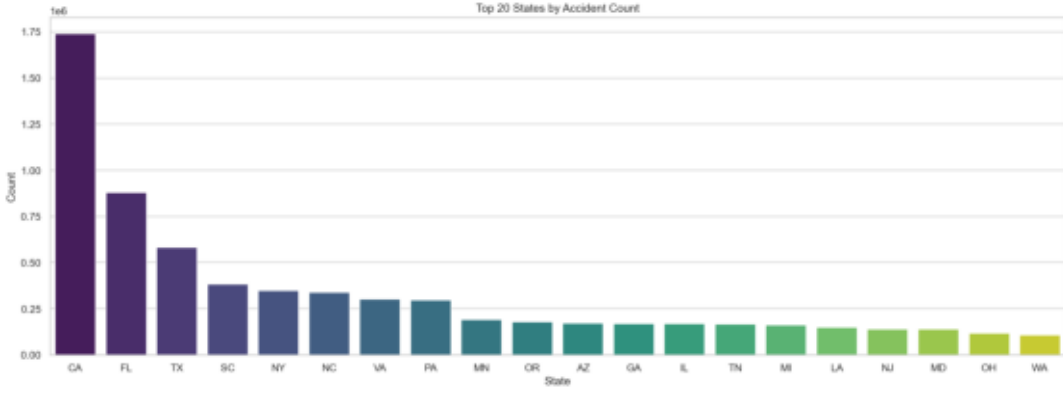
Eda Severity Distribution



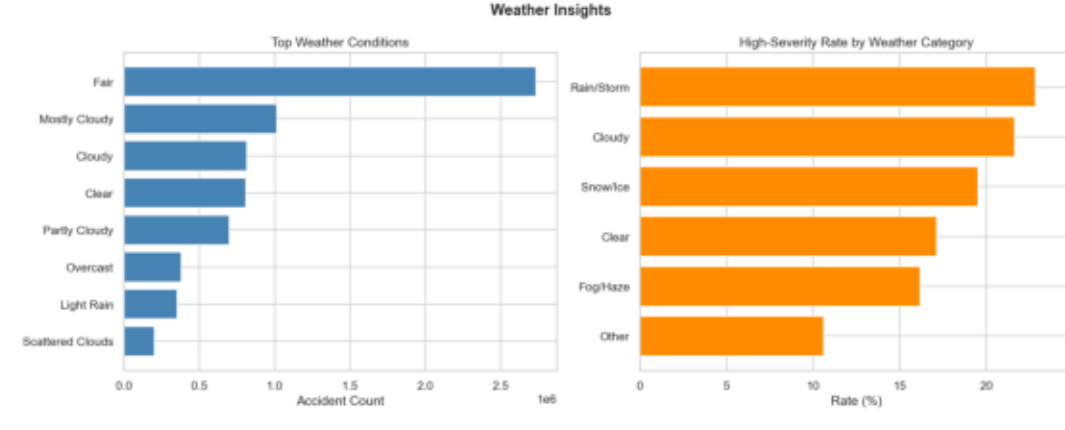
Eda Temporal Trends



Eda State Accidents

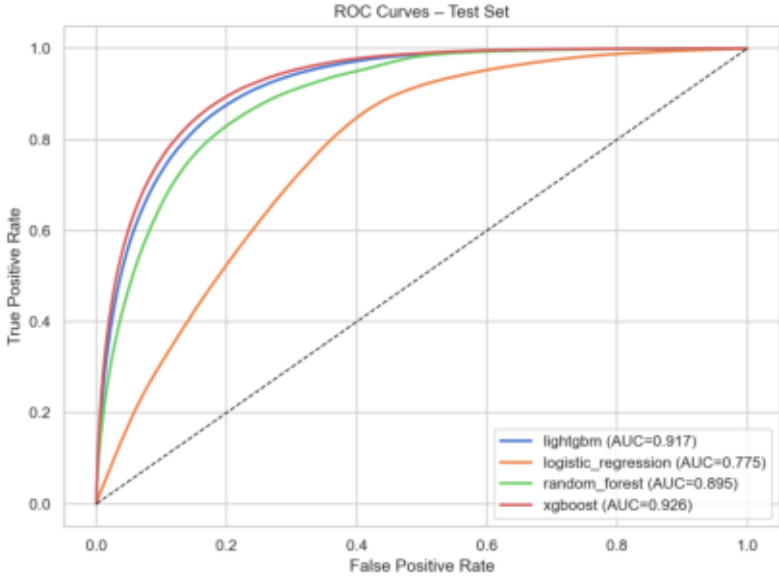


Eda Weather Insights

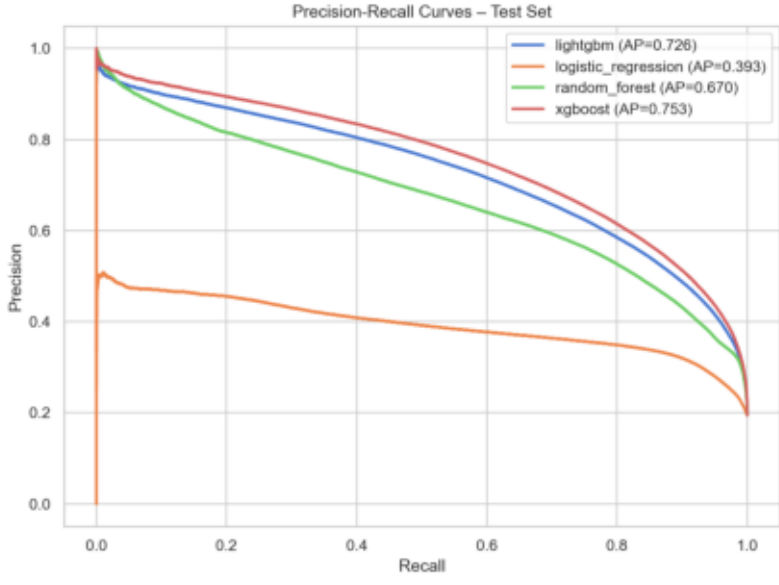


Model Evaluation Curves

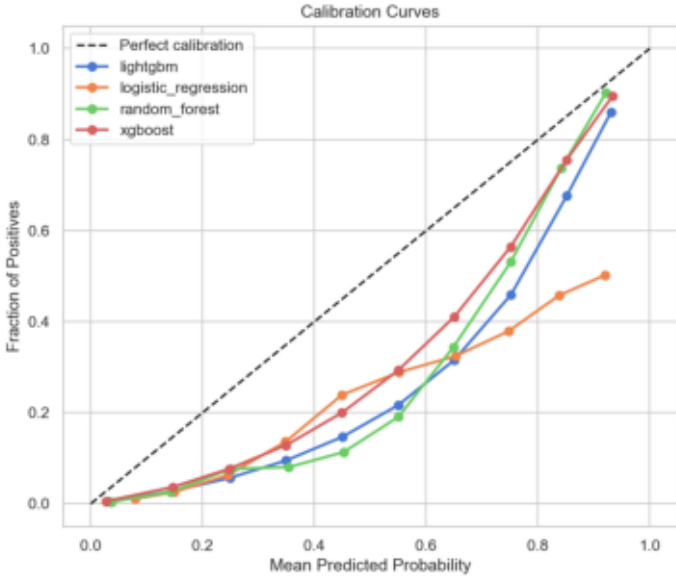
Roc Curves



Pr Curves

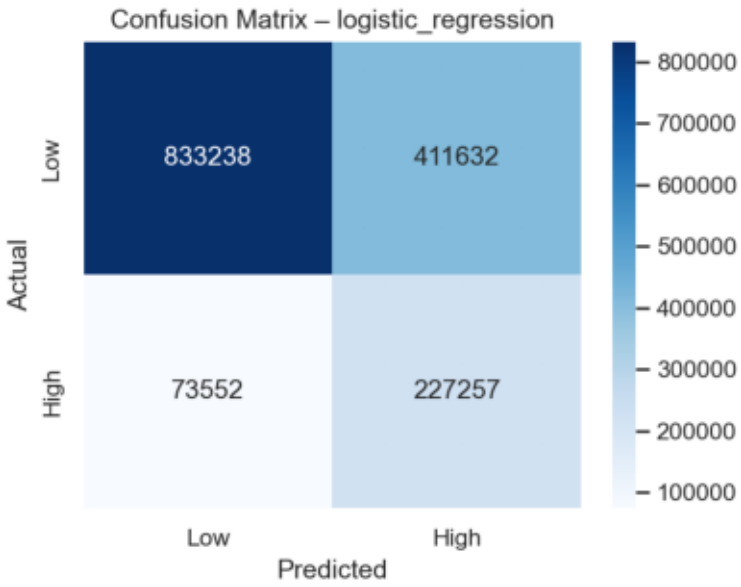


Calibration Curves

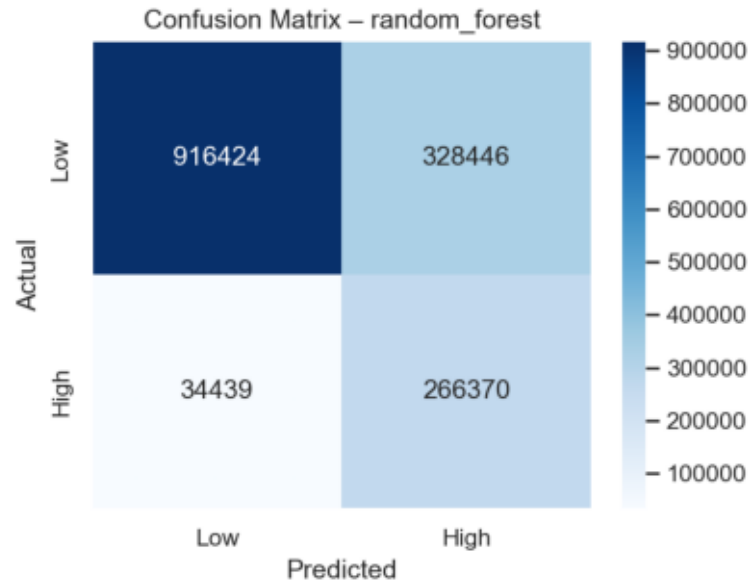


Confusion Matrices

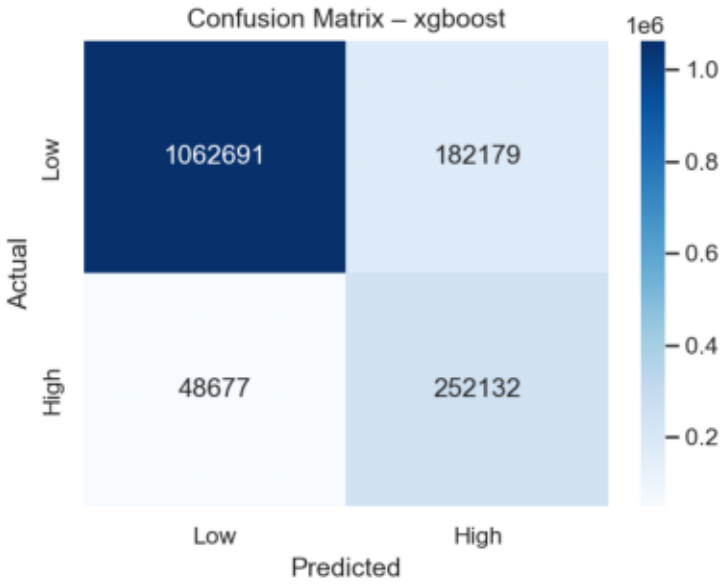
Cm Logistic Regression



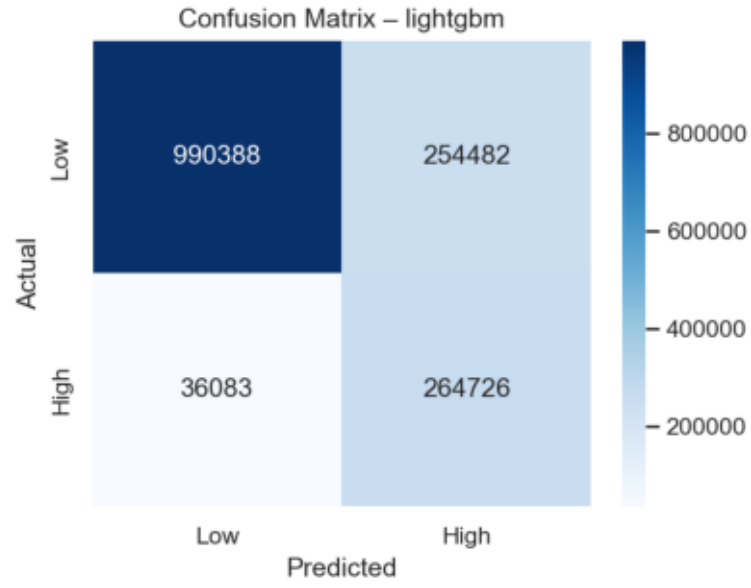
Cm Random Forest



Cm Xgboost



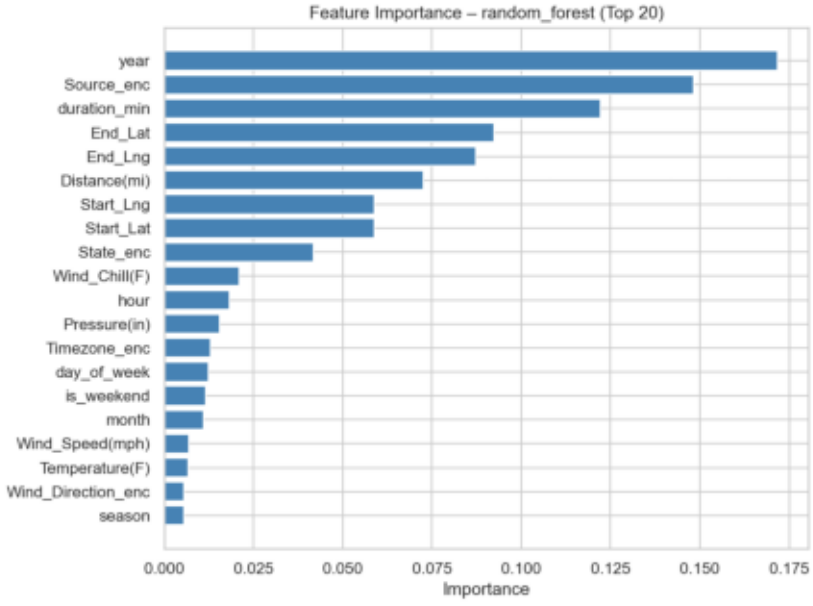
Cm Lightgbm



Feature Importance

Feat Imp Logistic Regression

Feat Imp Random Forest



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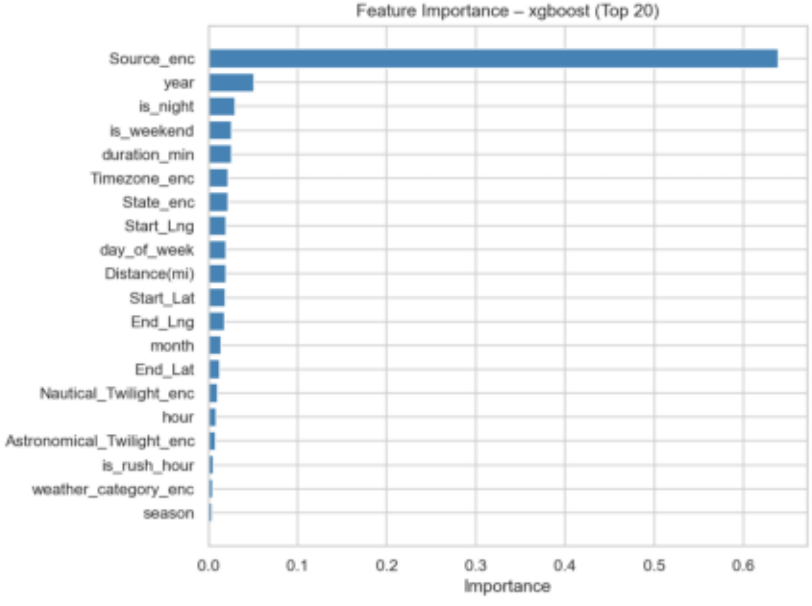
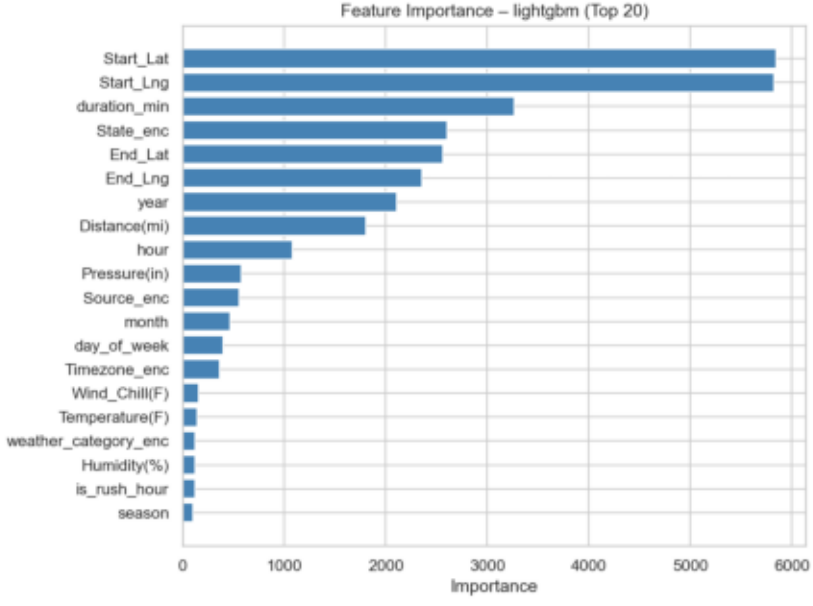
Feat Imp Lightgbm